



COMPETENCY STANDARD
FOR
PYTHON PROGRAMMING
(IT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **PYTHON PROGRAMMING (IT sector)** is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: <https://sicip.gov.bd/>

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of Occupation-specific industry experts of the industry sector, Consultants of SICIP, Program Officer(QA) to identify the competencies required of an occupation in the sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Shipbuilding industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (36 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-IT-PYP-01-G	Work in a team environment	1 Identify team goals and processes 2 Communicate and cooperate with team members 3 Work as a team member 4 Solve problems as a team member	12
SICIP-IT-PYP-02-G	Demonstrate Interpersonal and Communication Skills	1 Apply interpersonal skills at the workplace 2 Understand and communicate information and ideas 3 Prepare and present reports and workplace documents	12
SICIP-IT-PYP-03-G	Support Innovation and Manage Change	1 Identify needs for innovation in the workplace 2 Apply creative approach and solution 3 Support flexible and innovative ways of working 4 Adapt to emerging technological changes and opportunities	12
Total Hour			36

Sector Specific Competencies (24 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-IT-PYP-01-S	Access Information Using Internet and Electronic Mail	1. Access resources from internet 2. Use and manage electronic mail 3. Use audio/video tools for information transfer	12
SICIP-IT-PYP-02-S	Comply to Ethical Standards in IT Workplace	1. Uphold the requirements of clients 2. Deliver quality products and services 3. Maintain professionalism at workplace 4. Maintain workplace code of conduct	12
Total Hour			24

Occupation Specific Competencies (300 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-IT-PYP-01-O	Perform Basic Python Programming	1. Demonstrate knowledge on Basic Concepts of Python Programming 2. Install and Configure Python Environment 3. Install and Use Anaconda 4. Write and Execute Basic Python Scripts 5. Manipulate Strings in Python 6. Control Program Flow with Conditions and Loops 7. Manage Collections in Python 8. Define and Use Functions	80
SICIP-IT-PYP-02-O	Perform Advanced Python Programming	1. Implement Common Data Structures and Algorithms 2. Use Python Modules and Packages 3. Perform File Handling 4. Implement Object-Oriented Programming 5. Handle Python Exceptions	60
SICIP-IT-PYP-03-O	Develop Web Applications Using Django	1. Apply Basic Web Design Principles 2. Set Up Django Environment 3. Create Views, Templates, and Forms 4. Work with Models and Databases 5. Manage Django Admin Panel 6. Connect Django project to Github	100
SICIP-IT-PYP-04-O	Perform Data Analysis Using Python	1. Demonstrate Knowledge on Basic Statistical Concepts 2. Use Numpy for Data Manipulation 3. Work with Pandas for Data Analysis 4. Visualize Data with Matplotlib and Seaborn 5. Perform Descriptive Analysis Using Python	60
Total Hours			00

COMPETENCY STANDARD: PYTHON PROGRAMMING

Generic Competencies

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 12 hrs.	Unit Code: SICIP-IT-PYP-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Role and common goals of the team are defined from available <u>sources of information</u> . 1.3 Team structure, responsibilities and reporting relations are identified from team discussions and other external sources.
2. Communicate and cooperate with team members	1.4 Communication and negotiation skills are applied and maintained in all relevant situations. 1.5 Constructive contributions are made to <u>workplace discussions</u> on such issues as production, quality and safety. 1.6 Goals/ objectives and action plans undertaken in the workplace are communicated promptly. 1.7 Information regarding problems and issues are organized coherently to ensure clear and effective communication. 1.8 Dialogue is initiated with appropriate personnel. 1.9 Communication problems and issues are raised 1.10 Barriers to communication are identified and resolved
3. Work as a team member	1.1 Effective forms of communication are used to interact with <u>team members</u> in discussing team activities and objectives. 1.2 Mutual respect, empathy and active collaboration are demonstrated 1.3 Communication channels are followed as per <u>workplace context</u> .
4. Solve problems as a	1.1 Current and potential problems faced by team are

team member	identified. 1.2 Problems are investigated and analyzed. 1.3 Potential solutions of problem are identified. 1.4 Recommendations about possible solutions are developed, documented, ranked and presented to team members for decision.
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Range of variables:

Variables	Range (may include but not limited to)
1. Sources of information	1.1 Organizational structures 1.2 Operations Manuals 1.3 Job description 1.4 Standard operating procedures
2. Workplace discussions	1.1 Coordination meetings 1.2 Toolbox discussion 1.3 Peer-to-peer discussion
3. Team members	1.1 Coach / members 1.2 Supervisor / manager 1.3 Peers / colleagues 1.4 Other members /Employee representative of the organization.
4. Workplace context	1.1 National Laws and Statutes 1.2 Standard Operating Procedures 1.3 Workplace Rules and Regulations

Curricular Content Guide

1. Underpinning knowledge	1.1 Sources of information define 1.2 Team structure, role, and responsibility. 1.3 Individual member's roles and responsibilities. 1.4 Effective verbal communication methods 1.5 Communication flow and reporting structures. 1.6 Interpersonal communication skills. 1.7 Organization requirements for written and electronic communication methods 1.8 Communication problems and issues 1.9 Barriers in communication 1.10 Team planning. 1.11 Team meeting procedures. 1.12 Workplace etiquette 1.13 Industry maintenance, service and helpdesk practices, processes and procedures 1.14 Industry standard diagnostic tools 1.15 Malfunctions and resolutions
2. Underpinning skill	1.16 Organizing sources of information 1.17 Identifying the role and responsibility of the team.

	1.18 Identifying roles and responsibilities of individual members. 1.19 Identifying effective verbal communication methods 1.20 Identifying communication flow and reporting structure. 1.21 Identifying interpersonal communication skills 1.22 Complying with organization requirements for the use of written and electronic communication methods 1.23 Negotiation and communication skills 1.24 Participating in team discussion. 1.25 Working as a team member. 1.26 Participating in a variety of workplace discussions 1.27 Effective clarifying and probing skills 1.28 Identifying issues 1.29 Identifying current industry standard diagnostic tools 1.30 Describing common malfunctions and resolutions. 1.31 Determining the root cause of a routine malfunction
3. Underpinning Attitudes	1.32 Commitment to occupational health and safety practices 1.33 Promptness in carrying out activities 1.34 Tidiness and timeliness 1.35 Respect to peers, sub-ordinates and seniors in workplace 1.36 Environmental concern 1.37 Sincerity and honesty
4. Resource implications	The following resources must be provided: 1.38 Workplace (actual or simulated). 1.39 Relevant materials and equipment. 1.40 Relevant specifications or work instructions.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified team goals and processes. 1.2 communicated and cooperated with team members 1.3 worked as a team member 1.4 solved problems as a team member
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: DEMONSTRATE INTERPERSONAL AND COMMUNICATION SKILLS	Nominal Duration: 12 hrs.	Unit Code: SICIP-IT-PYP-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitude required to demonstrate interpersonal and communication skills. It specifically includes the tasks of applying interpersonal skills at the workplace, understanding and communicating information and ideas, preparing and presenting reports and workplace documents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply interpersonal skills at the workplace	1.1 <u>Workplace interpersonal skills</u> are interpreted. 1.2 The impacts of interpersonal skills are described. 1.3 Interpersonal skills are performed in accordance with workplace procedures. 1.4 <u>Ways of improving interpersonal skills</u> are identified and followed.
2. Understand and communicate information and ideas	2.1 <u>Communication tools and equipment</u> are operated, and faults are identified and reported. 2.2 Consultative processes is used to collect and convey information. 2.3 Information systems is used to store, retrieve and update information. 2.4 Information data and information are analyzed and interpreted. 2.5 Information is communicated with individuals and groups.
3. Prepare and present reports and workplace documents	3.1 <u>Workplace documents</u> are interpreted. 3.2 Appropriate format for document is used to meet workplace requirements. 3.3 Document is drafted in accordance with workplace procedures. 3.4 Document is edited and presented in a final version appropriate to task.

Range of variables:

Variables	Range (may include but not limited to)
1. Interpersonal skills	1.1 Effective communication 1.2 Cooperation 1.3 Giving and receiving feedback, feed-forward and

	professional criticism 1.4 Pressure handling 1.5 Time management 1.6 Acceptance of diversity 1.7 Supporting and encouraging others 1.8 Inclusiveness 1.9 Problem solving 1.10 Negotiation 1.11 Conflict management
2. Ways of improving interpersonal skills	1.1 Cultivate a positive outlook 1.2 Ask trusted friends or colleagues for constructive criticism 1.3 Control emotions 1.4 Observe other positive interpersonal interactions 1.5 Practice active listening 1.6 Seek out opportunities to build relationships 1.4 Practice empathy
3. Communication tools and equipment	1.1 Telephone 1.2 Internet 1.3 Mobile Phone 1.4 Fax machines 1.4 Two-way radio
4. Workplace documents	1.1 Business Letters 1.2 Business Reports 1.3 Transactional Documents 1.4 Memos 1.5 Forms 1.5.1 Memorandum 1.5.2 Requisitioning Form 1.5.3 Personnel Form 1.6 Safety Report Form

Curricular Content Guide

1. Underpinning knowledge	1.1 Interpersonal skills 1.2 Impacts of interpersonal skills 1.3 Consultative processes 1.4 Workplace documents
2. Underpinning skill	1.5 Demonstrating interpersonal skills 1.6 Collecting workplace information 1.7 Analyzing information and ideas 1.8 Preparing workplace documents 1.9 Presenting workplace documents
3. Underpinning Attitudes	1.10 Commitment to occupational health and safety practices 1.11 Promptness in carrying out activities 1.12 Tidiness and timeliness 1.13 Respect to peers, sub-ordinates and seniors in

	workplace 1.14 Environmental concern 1.15 Sincerity and honesty
4. Resource implications	The following resources must be provided: 1.1 Workplace (actual or simulated). 1.2 Relevant materials and equipment. 1.3 Relevant specifications or work instructions.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 applied interpersonal skills at the workplace 1.2 Communicated information and ideas 1.3 Prepared and presented reports and workplace documents
2. Methods of assessment	Competency should be assessed by: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: SUPPORT INNOVATION AND MANAGE CHANGE	Nominal Duration: 12 hrs.	Unit Code: SICIP-IT-PYP-03-G
Unit Descriptor: This unit covers the knowledge, skills, and attitudes required to support innovation and manage change. It specifically includes the tasks of identifying needs for innovation in the workplace, applying creative approaches and solution, supporting flexible and innovative ways of working and adapting to emerging technological changes and opportunities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify needs for innovation in the workplace	1.1. The need for <u>innovation</u> in scope of work is recognized 1.2. The value of <u>innovative practices</u> in the workplace is recognized 1.3. Existing way of working in the workplace are interpreted. 1.4. Drawbacks of existing way of working are identified. 1.5. Benefits of change are identified to make it consistent with organizational requirements 1.6. Realistic timelines and targets for implementation of changes are set.
2. Apply creative approach and solution	2.1. Opportunities and creative approaches to implement innovation practices are identified 2.2. Creative approaches of coworkers pertaining to work practices are analyzed and incorporated 2.3. Innovation in accordance with organizational requirements changes are implemented
3. Support flexible and innovative ways of working	3.1. Maximum innovative opportunities are promoted. 3.2. Work assignments to facilitate innovative work skills are organized. 3.3. Team members are provided with guidance and coaching on innovation in the workplace. 3.4. Models of innovative work practice are provided and discussed 3.5. Appropriate environment for learning and innovation is maintained.
4. Adapt to emerging technological changes and opportunities	4.1. Usages of different <u>technologies</u> is determined based on job requirements. 4.2. Appropriate technology is selected as per work specifications. 4.3. Relevant technology is effectively used in carrying out functions.

	4.4. Appropriate implementation tools are used as per task requirement.
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Range of variables:

Variables	Range (may include but not limited to)
1. Innovative practices	1.1. Self-directed support to implement new ideas 1.2. Collaborative arrangement to apply new method of working 1.3. Making scope of work more efficient to use new technology
2. Innovation	1.4. New ideas 1.5. Different ideas 1.6. New methods of doing work 1.7. Use of new Technology
3. Technologies	1.8. Office technology 1.9. Industrial technology 1.10. System technology 1.11. Information technology 1.12. Training technology

Curricular Content Guide

1. Underpinning knowledge	1.1. Definition of innovation 1.2. Current practice in own scope of work 1.3. Workplace procedures 1.4. Support required to generate creative ideas 1.5. Difference between innovation and creativity 1.6. Common effects of change and innovation in the workplace 1.7. Industrial and organizational context of change 1.8. Organization's policies, plans, procedures and structure 1.9. Knowledge of resources required by the organization's operations 1.10. Awareness on technology and its function 1.11. Operating instructions 1.12. Applicable software 1.13. Company policy in relation to relevant technology 1.14. Technology adaptability
2. Underpinning skill	1.1. Identifying resources required for creativity and innovation 1.2. Contributing to brainstorming session 1.3. Identifying issues and concerns of one's scope of work 1.4. Encouraging co-workers to generate and develop ideas 1.5. Evaluating potential obstacles to and opportunities for creativity and innovation 1.6. Sharing of best practices related to innovation and creativity

	1.7. Applying relevant technology 1.8. Using update machine and Software 1.9. Acquiring troubleshooting skills Identifying issues 1.10. Identifying current industry standard diagnostic tools 1.11. Describing common malfunctions and resolutions. 1.12. Determining the root cause of a routine malfunction
3. Underpinning Attitudes	1.13. Commitment to occupational health and safety 1.14. Promptness in carrying out activities 1.15. Sincere and honest to duties 1.16. Eagerness to learn 1.17. Tidiness and timeliness 1.18. Environmental concerns 1.19. Respect for rights of peers and seniors at workplace 1.20. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.21. Workplace (actual or simulated). 1.22. Relevant materials and equipment. 1.23. Relevant specifications or work instructions.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified need for innovation in the area of work 1.2 applied creative approach and solution 1.3 supported flexible and innovative ways of working 1.4 adapted to emerging technological changes and opportunities
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: ACCESS INFORMATION USING INTERNET AND ELECTRONIC MAIL	Nominal Duration: 12 hrs.	Unit Code: SICIP-IT-PYP-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitude required to access information using internet and electronic mail. It specifically includes the tasks of accessing resources from internet, using and managing electronic mail, and using audio/video tools for information transfer.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Access resources from internet	1.1 Appropriate internet <u>browsers</u> are selected and installed. 1.2 Internet browser is opened and web address / URL is written/selected in /from address bar to access <u>information.</u> 1.3 <u>Search engines</u> are used to access information 1.4 Video / Information are Shared /downloaded / uploaded from / to web site/ <u>social media.</u> 1.5 <u>Web based resources</u> are used. 1.6 Etiquette (or web etiquette) principles are searched and followed.
2. Use and manage electronic mail	2.1 <u>Email services</u> are identified and selected to create a new email address 2.2 Mail account is created. 2.3 Document is prepared, attached and sent to different types of recipient. 2.4 Mail is read, forwarded, replied and deleted as per requirement. 2.5 Custom email folders are created and manipulated. 2.6 Mail message is printed.
2. Use audio/video tools for information transfer	2.1 Audio and video tools are identified 2.2 Apps using audio/video tools are identified 2.3 Information is transferred with apps using audio/video tools

Range of Variables

Variable	Range
	May include but not limited to:
1. Browsers	1.1 Mozilla Firefox

	1.2 Google chrome 1.3 Internet explorer 1.4 Opera
2. Information	2.1. Text information 2.2. Graphics 2.3. Video
3. Search engines	3.1. Google 3.2. Yahoo 3.3. AltaVista 3.4. Msn 3.5. Bing
4. Social media.	4.1 Face book 4.2 Twitter 4.3 LinkedIn 4.4 YouTube
5. Web based services	5.1 Drive 5.2 Calendar 5.3 Map 5.4 Translator 5.5 Docs 5.6 Search
6. Email services	6.1 Free mail services –Gmail, Yahoo, Hotmail 6.2 Web mail services.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Internet 1.2 www 1.3 web site 1.4 web address 1.5 URL 1.6 Web browsers 1.7 Search engines 1.8 Information 1.9 Social media 1.10 Web based services 1.11 Folder manipulation
2. Underpinning Skills	2.1 Accessing and sharing resources from internet 2.2 Downloading /uploading file, documents and video from /to web sites 2.3 Sending and receiving mail through mail service. 2.4 Using audio/video tools to share information
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn

	3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Relevant tools, Equipment, software and facilities needed to perform the activities. 4.2 Required learning materials.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 accessed resources from internet. 1.2 used and managed electronic mail 1.3 used audio/video tools for information transfer
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: COMPLY TO ETHICAL STANDARDS IN IT WORKPLACE	Nominal Duration: 12 hrs.	Unit Code: SICIP-IT-PYP-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to comply to ethical standards in IT workplace. It specifically includes the tasks of upholding the requirements of clients, delivering quality products and services, maintaining professionalism at workplace and maintaining workplace code of conduct.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Uphold the requirements of clients	1.1 Clients' requirements are identified. 1.2 Confidentiality of information is maintained in accordance with workplace policies/organizational policies/ <u>national legislation</u> . 1.3 Potential conflicts of interest are identified and involved parties of potential conflicts are notified. 1.4 Proprietary rights of client/customer are asserted.
2. Deliver quality products and services	2.1 Products and services are provided according to the clients' requirements. 2.2 Work is completed as per standards. 2.3 Quality processes are implemented when developing products and services.
3. Maintain professionalism at workplace	3.1 Work processes are delivered as per standards. 3.2 Skills, knowledge and qualifications are presented in a professional manner. 3.3 Services and products developed by self and others are delivered as per workplace standard. 3.4 Unbiased and objective information are provided to clients. 3.5 Realistic estimates for time, cost and delivery of outputs are presented during negotiation.
4. Maintain workplace code of conduct.	4.1 Workplace code of conduct are interpreted 4.2 Workplace code of conduct is followed.

Range of Variables

Variable	Range May include but not limited to:
1. National legislation	1.1 Industry/sector code of ethics 1.2 International and national guidelines for customer protection 1.3 International and national copyright laws

	1.4 Occupational health and safety requirements 1.5 Intellectual property rights law 1.6 Legal and regulatory policies in the information technology sector 1.7 Fire safety rules and regulations
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Corporate code of confidentiality of information organizational policies, national legislation and workplace policies in relation to IT sector 1.2 Law and regulations pertaining to proprietary rights 1.3 Quality processes for products and services 1.4 Procedure of provided to client information 1.5 Method of estimating for time, cost and delivery products and services 1.6 Workplace code of conduct in IT sector
2. Underpinning Skills	2.1 Upholding confidentiality of information in accordance with organizational policies, national legislation and workplace policies 2.2 Asserting proprietary rights of client/customer 2.3 Completing work in accordance with industry and international standards 2.4 Implementing quality processes when developing products and services 2.5 Delivering correctly services and products developed by self and others 2.6 Providing unbiased and objective information are to clients. 2.7 Presenting realistic estimates for time, cost and delivery of outputs during negotiation 2.8 Following workplace code of conduct
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Environmental concerns 3.3 Eagerness to learn 3.4 Tidiness and timeliness 3.5 Concern to proper use of tools 3.6 Orderliness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Work place code of conduct 4.3 Papers 4.4 Work books

Assessment Evidence Guide

1. Critical Aspects of	Assessment required evidence that the candidate:
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Competency	1.1 uphold the requirements of clients. 1.2 delivered quality products and services 1.3 maintained professionalism at workplace 1.4 maintained workplace code of conduct.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: PERFORM BASIC PYTHON PROGRAMMING	Nominal Duration: 80 Hrs.	Unit Code: SICIP-IT-PYP-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform basic Python programming. It specifically includes the tasks of demonstrating knowledge on Basic Concepts of Python Programming, installing and configuring Python environment, installing and using Anaconda, writing and executing basic Python scripts, manipulating strings in Python, controlling program flow with conditions and loops, managing collections in Python, and defining and using Functions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Demonstrate knowledge of the Basic Concepts of Python Programming	1.1 The history and importance of Python programming language are described. 1.2 Various <u>application fields of Python</u> are identified. 1.3 Differences between compilers and interpreters are explained. 1.4 Key features of Python is listed. 1.5 The concept of the Python and Python code is explained.
2. Install and Configure Python Environment	2.1 Python installation methods for various <u>operating systems</u> are identified and executed. 2.2 Environment variables and path settings are checked and configured. 2.3 Python packages are installed and managed using pip. 2.4 Dependency conflicts are identified and resolved through the use of virtual environments. 2.5 The installation of Python IDEs and text editors is completed. 2.6 The Python environment is updated regularly to ensure compatibility with the latest features and security patches.
3. Install and Use Anaconda	3.1 Anaconda is downloaded and installed ensuring compatibility with the operating system. 3.2 JupyterLab is started and configured, setting up an interactive environment for python programming and data analysis. 3.3 Python programs are written and executed using the selected <u>Python IDEs</u> and <u>Interactive computing environments</u> . 3.4 Common IDE issues are troubleshoot to ensure a smooth coding experience.

	3.5 Package management is performed in IDEs, Jupyter Notebook and Google Colab. 3.6 Jupyter Notebook features for documentation and code execution are demonstrated. 3.7 Anaconda navigator is used to manage environments.
4. Write and Execute Basic Python Scripts	4.1 Basic scripts are created with correct syntax. 4.2 Python indentation rules are followed. 4.3 Comments are included to explain the code. 4.4 Variables are named using standard conventions. 4.5 Data types are assigned and typecasting is performed. 4.6 Operators are applied in correct precedence. 4.7 Boolean expressions are evaluated. 4.8 Error handling is demonstrated for managing exceptions.
5. Manipulate Strings in Python	5.1 String slicing is performed to extract substrings. 5.2 Strings are concatenated and formatted. 5.3 Escape characters are used for special formatting. 5.4 Built-in string methods are used to modify and inspect strings.
6. Control Program Flow with Conditions and Loops	6.1 Conditional statements are used for branching in program flow. 6.2 The usage of case statements is demonstrated. 6.3 Loops are used for repeated execution of code blocks. 6.4 Break, continue, and pass statements are applied to control the flow within loops. 6.5 Nested loops and conditions are implemented. 6.6 Logical operators are used to combine multiple conditions effectively. 6.7 Infinite loops and their safe termination are demonstrated.
7. Manage Collections in Python	7.1 Python collections are created. 7.2 The difference between lists, tuples and sets are explained. 7.3 List, set and dictionary comprehensions are used. 7.4 The concept of a frozen set is explained. 7.5 Built-in methods are applied to handle set operations. 7.6 List operations are performed. 7.7 Tuple operations are performed. 7.8 Set operations are performed. 7.9 Dictionary operations are performed.
8. Define and Use Functions	8.1 Functions are declared with parameters. 8.2 Functions are called from the main code. 8.3 Return values are captured and stored for further processing. 8.4 Arbitrary arguments, keyword arguments, and recursion are demonstrated. 8.5 Lambda functions are created for concise operations.

	8.6 Default parameters values are assigned to functions. 8.7 The scope of variables is explained to clarify their usage within functions.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Application fields of Python	1.1 Web development 1.2 Data analysis 1.3 Artificial intelligence 1.4 Scientific computing 1.5 Automation 1.6 Game development 1.7 Networking
2. Operating systems	2.1 Windows 2.2 Linux 2.3 MacOS
3. Python IDEs	3.1 PyCharm 3.2 VSCode
4. Interactive computing environment	4.1 Jupyter Notebook 4.2 Google Colab
5. Package Management	5.1 Pip 5.2 Conda 5.3 virtual environments
6. Basic Scripts	6.1 Small programs focusing on input output statements or interactive testing
7. Variables	7.1 Local 7.2 Global 7.3 Dynamically typed
8. Conditional Statements	8.1 if, if-else, if-elif-else
9. Loops	9.1 For, while
10. Operators	10.1 Arithmetic 10.2 Assignment 10.3 Comparison 10.4 Logical 10.5 Bitwise
11. Built-in string methods	11.1 Capitalize 11.2 Split 11.3 Replace 11.4 Rjust 11.5 Casefold 11.6 Count 11.7 Encode 11.8 Endswith 11.9 Expandtabs

	11.10 Find 11.11 Index 11.12 Isalphanumeric 11.13 Isalpha 11.14 Isascii 11.15 Isdecimal
12. List operations	12.1 Manually - access, change, add, remove, sort, copy, join, slicing 12.2 Methods - append, clear, copy, count, extend, insert, pop, remove, reverse
13. Tuple Operations	13.1 Manually - access, len, min, max 13.2 Methods – count, index
14. Set operations	14.1 Manually - add, remove, access, join, loop; 14.2 Methods - union, intersection, add, difference, discard, isdisjoint, issubset, issuperset, symmetric difference
15. Dictionary Operations	15.1 Manually - access, change, add, remove, copy, join; 15.2 Methods - clear, fromkey, get, items, values, keys. Pop, popitems, update

Curricular Content Guide

1. Underpinning Knowledge	1.1 History, applications, and importance of Python 1.2 Differences between interpreters and compilers 1.3 Structure of Python scripts, indentation rules, and commenting 1.4 Naming conventions and data types for variables. 1.5 Operator categories and precedence. 1.6 Function definitions, parameters, return statements, and recursion.
2. Underpinning Skills	2.1 Installing Python on different operating systems and set environment variables. 2.2 Using different IDEs to write, run, and debug Python code 2.3 Installing Python packages and managing dependencies using virtual environments 2.4 Utilizing Anaconda to manage packages and create environments 2.5 Operating Jupyter Notebook and Google Colab to produce interactive code and documentation 2.6 Writing Python scripts using correct syntax & indentation through operators and variables 2.7 Manipulating strings with slicing, formatting, and built-in methods. 2.8 Implementing conditional statements and loops for program flow. 2.9 Creating and modifying lists, tuples, dictionaries, and sets. 2.10 Defining and calling functions with arguments and

	return values. 2.11 Using recursion and lambda functions
3. Underpinning Attitudes	3.1 Creativity and attention to detail 3.2 Eagerness to experiment and iterate 3.3 Patience and precision in design work 3.4 Commitment to maintaining high design standards
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer lab 4.3 Desktop PCs

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 demonstrated knowledge of the Basic Concepts of Python Programming 1.2 installed and configured Python Environment 1.3 installed and used Anaconda 1.4 write and executed basic Python Scripts 1.5 manipulated strings in Python 1.6 controlled program flow with conditions and loops 1.7 managed collections in Python 1.8 defined and used functions
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ADVANCED PYTHON PROGRAMMING	Nominal Duration: 60 Hrs.	Unit Code: SICIP-IT-PYP-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform advanced Python programming. It specifically includes the tasks of implementing common data structures and algorithms, using Python modules and packages, performing file handling, implementing object-oriented programming, and handling Python exceptions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Implement Common Data Structures and Algorithms	1.1 Common <u>data structures</u> are identified and explained. 1.2 <u>Searching and sorting algorithms</u> are described. 1.3 Data structures are implemented in Python. 1.4 Algorithms are tested and verified for correctness. 1.5 Time and space complexity of algorithms are analyzed to access efficiency. 1.6 The appropriate data structure is chosen based on the problem requirements.
2. Use Python Modules and Packages	2.1 Basic concepts of modules and packages are outlined. 2.2 <u>Built-in modules</u> are used for common tasks. 2.3 Usage of <u>networking modules</u> are demonstrated. 2.4 The tkinter module is used for basic GUI applications. 2.5 External packages are installed and managed. 2.6 Custom modules and packages are developed. 2.7 Module documentation and versioning are applied to maintain clarity and track changes.
3. Perform File Handling	3.1 File creation, opening, reading, writing, and closing operations are performed. 3.2 File paths and directories are manipulated using <u>sys, os and shutil modules</u> . 3.3 <u>File compression</u> and <u>encoding</u> techniques are demonstrated. 3.4 <u>File permissions</u> and encoding techniques are managed using relevant functions and methods. 3.5 File content is searched and manipulated using built-in methods.
4. Implement Object-Oriented Programming	4.1 <u>Basic concepts of Object Oriented Programming (OOP)</u> are explained. 4.2 Classes, objects, methods, attributes, and constructors are created. 4.3 <u>Inheritance</u> is demonstrated, and overriding and super functions are explained. 4.4 <u>Encapsulation</u> and <u>polymorphism</u> concepts are

	applied. 4.5 Abstraction method is explained. 4.6 Abstract base classes and interface classes are created. 4.7 The concept of method resolution order (MRO) and multiple inheritance is demonstrated.
5. Handle Python Exceptions	5.1 Error handling and exceptions are explained to ensure robust program execution. 5.2 The try-except block is used to manage errors allowing for the execution of code with error handling mechanism is place. 5.3 Multiple exceptions and a single block for multiple exceptions are demonstrated. 5.4 Else and finally blocks are demonstrated. 5.5 Built-in exceptions are identified and explained for efficient error management. 5.6 Custom exceptions are raised where needed. 5.7 Exception chaining and nested exception are demonstrated.

Range of Variables

Variable	Range (Includes but not limited to):
1. Data Structures	1.1 Arrays 1.2 Stacks 1.3 Queues 1.4 Linked lists 1.5 Trees and graphs
2. Searching and Sorting algorithms	2.1 DFS 2.2 BFS 2.3 Divide and conquer 2.4 Greedy 2.5 Dynamic
3. Built-in modules	3.1 Sys 3.2 Os 3.3 Math 3.4 Datetime
4. Networking modules	4.1 Requests 4.2 Http 4.3 Socket 4.4 Email
5. Sys, os and shutil modules	5.1 Sys - sys.path, sys.version, sys.platform, sys.getsizeof(), sys.executable 5.2 os - os.getcwd(), os.chdir(path), os.mkdir(path), os.remove(path), os.rmdir(path), os.rename(src, dst), os.path.exists(path), os.path.isfile(path), os.path.isdir(path), os.path.join(path1, path2),

		os.walk() 5.3 shutil.copy(src, dst), shutil.move(src, dst)
6. File comprehension	6.1	File compression modules - gzip, zipfile, tarfile, rarfile, py7zr
7. Encoding	7.1	Encoding modules - codecs, io, locale
8. File permission	8.1	File Permission functions - os.chmod(path, mode), os.access(path, mode), Path(path).chmod(mode), Path(path).stat().st_mode
9. Basic concepts of Object Oriented Programming	9.1	Classes, objects, methods, attributes, and constructors
10. Inheritance	10.1	Single, multiple, multilevel, hierarchical
11. Encapsulation	11.1	Private and protected members ; Getter and Setter Methods
12. Polymorphism	12.1	Method overriding, operator overloading
13. Built-in exceptions	13.1 13.2 13.3 13.4 13.5 13.6 13.7 13.8	ZeroDivisionError: Division by zero. TypeError: Invalid operation on a data type. ValueError: Invalid value. FileNotFoundError: File not found. IndexError: Index out of range. KeyError: Key not found in dictionary. AttributeError: Attribute not found. ImportError: Module import failed

Curricular Content Guide

1. Underpinning Knowledge	1.1. Common data structures (array, stack, queue, linked list, tree, graph) and related algorithms. 1.2. Python modules, packages, and standard library functions. 1.3. File handling concepts, path manipulations, compression, and encoding. 1.4. Object-oriented programming principles (inheritance, polymorphism, encapsulation, abstraction). 1.5. Exception handling flow (try-except, else, finally) and built-in exception types.
2. Underpinning Skills	2.1. Implementing arrays, stacks, queues, trees, and graphs in Python. 2.2. Applying searching and sorting algorithms. 2.3. Using Python modules (sys, os, math, datetime, etc.) and installing external packages. 2.4. Performing file handling tasks (read, write, compress, encode, set permissions). 2.5. Designing classes and objects with inheritance, polymorphism, and encapsulation. 2.6. Handling exceptions with try-except blocks and raising custom exceptions.
3. Underpinning Attitudes	3.1 Curiosity and willingness to learn

	3.2 User-focused problem-solving mindset 3.3 Attention to detail in design and testing 3.4 Adaptability to industry trends and feedback
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer lab 4.3 Desktop PCs

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 implemented common data structures and algorithms. 1.2 used Python modules and external packages. 1.3 performed file handling. 1.4 implemented Object-Oriented programming. 1.5 handled Python exceptions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP WEB APPLICATIONS USING DJANGO	Nominal Duration: 100 Hrs.	Unit Code: SICIP-IT-PYP-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to develop web applications using Django. It specifically includes the tasks of applying basic web design principles, setting up Django environment, creating Views, Templates and Forms, working with models & databases, managing Django Admin panel, and connecting Django project to Github.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply Basic Web Design Principles	1.1 Major <u>Components of webpage</u> are explained. 1.2 Basic concepts of front-end and back-end development are explained. 1.3 <u>HTML structures</u> are demonstrated. CSS is applied using inline, internal, and external methods. 1.4 JavaScript (JS) is demonstrated for simple functions, conditions, and loops. 1.5 Common <u>JavaScript frameworks</u> are explained. 1.6 Responsive web design principles are applied.
2. Set Up Django Environment	1.7 Django is installed using package managers and its installation is verified. 1.8 Virtual environments are created and activated to isolated project dependencies. 1.9 A new Django project is created and run on the localhost to ensure proper setup. 1.10 Django apps are created and registered within the project. 1.11 The database is set up and migrated using Django's built-in-migration system. 1.12 Basic Django routing is configured to ensure the project can handle HTTP requests.
3. Create Views, Templates, and Forms	1.13 <u>Views</u> are defined and displayed in the browser to show dynamic content. 1.14 URL routing is configured to link views and templates using Django's file. 1.15 HTML templates are integrated to render dynamic content. 1.16 Differences between model view controller (MVC) and model view template (MVT) are explained. 1.17 Forms are created for user input and authentication. 1.18 Multiple templates are managed and extended. 1.19 Template filters and tags are used to manipulate and

	display.
4. Work with Models and Databases	1.20 Models are defined with fields, relationships, and methods. 1.21 Migrations are performed to create or update database schema. 1.22 CRUD operations are carried out using Django's ORM. 1.23 Forms are linked to models for data handling to facilitate form creation and data validation. 1.24 Model relationships are implemented to link related data across different models. 1.25 Database queries are applied using Django's query sets. 1.26 Custom model methods and manager methods are defined to implement specific business logic.
5. Manage Django Admin Panel	1.27 The admin panel is enabled and configured. 1.28 Models are registered for administrative access. 1.29 Admin views are customized to suit project requirements. 1.30 Permissions and user actions are defined for secure management. 1.31 Filters are used in admin panel to allow users to quickly search record. 1.32 The admin interface is extended with custom actions. 1.33 The appearance of the admin panel is customized using custom templates and static files.
6. Connect Django project to Github	1.34 Version controlling and its benefits are explained. 1.35 A Github account is set up and a new repository is created to store and manage. 1.36 Git is initiated in the Django project. 1.37 Files are added to the git repository using to stage changes for committing. 1.38 Local repository is connected to Github and the project is pushed. 1.39 Git branches are used to manage different features or fixes.

Range of Variables

Variable	Range (Includes but not limited to):
1. Components of webpage	1.1 Header 1.2 Navigation Menu 1.3 Footer 1.4 Main Content Area 1.5 Call-to-Action (CTA) Buttons 1.6 Sidebar 1.7 Images and Graphics

	1.8 Forms 1.9 Search Bar 1.10 Social Media Links
2. HTML structures	2.1 Headers 2.2 Paragraphs 2.3 Lists 2.4 Forms 2.5 Tables 2.6 Images
3. JS frameworks	3.1 Bootstrap 3.2 React 3.3 Vue.js, 3.4 Node.js
4. Views	4.1 Function-based views 4.2 Class-based views 4.3 API based views
5. CRUD operations	5.1 Create, 5.2 Read 5.3 Update 5.4 Delete

Curricular Content Guide

1. Underpinning Knowledge	1.1. Basic HTML, CSS, and JavaScript concepts. 1.2. Front-end vs. back-end roles in web development. 1.3. Django project structure (models, views, templates, URLs, admin). 1.4. Database interactions through Django's ORM. 1.5. Admin panel configuration and user permission management. 1.6. Basic git commands
2. Underpinning Skills	2.1. Applying HTML and CSS to create basic page layouts. 2.2. Using JavaScript for simple dynamic behavior. 2.3. Installing Django 2.4. Configuring Django projects, apps, and URLs. 2.5. Defining views, templates, and forms for user interaction 2.6. Building models and performing CRUD operations on databases 2.7. Managing the admin panel for model administration and permissions. 2.8. Pushing Django project on Github
3. Underpinning Attitudes	3.1 Analytical thinking and problem-solving mindset 3.2 User-centered approach to requirement gathering 3.3 Attention to detail in research and analysis 3.4 Pro-activeness in identifying feasibility challenges
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment required for the job

	4.3 Materials required 4.4 Work books 4.5 Diagram and instruction manual
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied basic web design principles. 1.2 set up Django environment. 1.3 created views, templates, and forms. 1.4 worked with models and databases. 1.5 managed Django admin panel. 1.6 connected Django project to Github.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM DATA ANALYSIS USING PYTHON	Nominal Duration: 60 Hrs.	Unit Code: SICIP-IT-PYP-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform data analysis using Python. It specifically includes the tasks of demonstrating knowledge on basic statistical concepts, using Numpy for data manipulation, working with Pandas for data analysis, visualizing data with Matplotlib and Seaborn, and performing descriptive analysis using Python.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Demonstrate Knowledge on Basic Statistical Concepts	1.1 <u>Measures of central tendency</u> and <u>dispersion</u> are explained. 1.2 Frequency distributions are interpreted. 1.3 Dependent and independent variables are explained. 1.4 Correlation is explained to determine the strength. 1.5 Regression is interpreted to model and predict the relationship between a dependent variable.
2. Use Numpy for Data Manipulation	2.1 Numpy arrays are created using linspace and arange. 2.2 Reshaping, indexing, and slicing operations are performed. 2.3 Array arithmetic operations are applied to perform calculations on arrays. 2.4 Numpy broadcasting is used to perform operations on arrays of different shapes. 2.5 Statistical functions are applied to calculate summary statistics of Numpy arrays. 2.6 Numpy functions for handling missing are demonstrated. 2.7 Numpy random number generation functions are used.
3. Work with Pandas for Data Analysis	3.1 Pandas Series and DataFrames are created and managed. 3.2 <u>Data cleaning and preprocessing tasks</u> are performed. 3.3 Time handling is demonstrated using Pandas. 3.4 Basic visualization using Pandas is demonstrated. 3.5 Filtering and selecting data are applied based on conditions.
4. Visualize Data with Matplotlib and Seaborn	4.1 <u>Plots</u> are generated with Matplotlib's object-oriented approach. 4.2 Multiple subplots are created for comparative visualization. 4.3 Seaborn is used for aesthetics.
5. Perform Descriptive	5.1 Mean, median, mode, and standard deviation, and

Analysis Using Python	variance are computed. 5.2 Frequency distributions are visualized. 5.3 Correlation and simple regression analyses are conducted.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Measures of central tendency	1.1 Mean 1.2 Median 1.3 Mode
2. Dispersion	2.1 Range 2.2 Variance 2.3 Standard Deviation 2.4 Interquartile Range
3. Data cleaning and preprocessing tasks	3.1 Handling Missing Data 3.2 Removing Duplicates 3.3 Data Type Conversion 3.4 Renaming Columns 3.5 Reordering Columns 3.6 Filtering Data 3.7 Sorting Data 3.8 Merging and Joining Data 3.9 Grouping and Aggregation 3.10 Handling Indexes 3.11 Dropping Columns 3.12 Adding New Columns 3.13 Saving Data
4. Plots	4.1 Line 4.2 Bar 4.3 Scatter 4.4 Histogram 4.5 Pie chart 4.6 Stacked bar 4.7 Distribution plots 4.8 Categorical plots 4.9 Heat maps

Curricular Content Guide

1. Underpinning Knowledge	1.1. Measures of central tendency, variance, and frequency distributions. 1.2. Numpy for array manipulation and mathematical operations. 1.3. Pandas for data cleaning, preprocessing, and manipulation.
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	1.4. Matplotlib and Seaborn for data visualization. 1.5. Descriptive analysis, correlation and simple regression
2. Underpinning Skills	2.1. Identifying relevant statistical measures for data analysis. 2.2. Using Numpy arrays for mathematical operations. 2.3. Handling DataFrames with Pandas for data manipulation. 2.4. Visualizing data with Matplotlib and Seaborn. 2.5. Performing descriptive analyses (mean, median, mode, standard deviation, correlation, regression).
3. Underpinning Attitudes	3.1 Attention to detail in wireframe design 3.2 User-centered approach to structuring interfaces 3.3 Willingness to iterate and improve based on feedback 3.4 Commitment to following design best practices
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Materials, tools and equipment needed for the activity 4.3 Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 demonstrated knowledge on basic statistical concepts. 1.2 used Numpy for data manipulation. 1.3 worked with Pandas for data analysis. 1.4 visualized data with matplotlib and seaborn. 1.5 performed descriptive analysis using Python.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Python Programming (IT Sector)
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

SI. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops with internet facility	26
2.	Multimedia projector & screen	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Python Programming (IT Sector), the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.